Lesson Planning

Complete Lesson Plan Book

Class: Class 12 – FSc Pre-Engineering

Subject: Physics

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

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Unit 1: Electrostatics

Learning Objectives

• Define and explain Coulomb's law in the context of Electrostatics.
• Differentiate between electric field and electric potential with suitable examples.
• Solve numerical problems related to capacitors accurately.
• Apply the ideas of Electrostatics to everyday situations involving Gauss's law.

Key Concepts

• Meaning, definition and scope of Coulomb's law.
• Important properties, types and examples of electric field.
• Relationship between electric potential and capacitors.
• Practical applications of Electrostatics, especially Gauss's law.
• Key terms and a summary of the main points of Electrostatics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Electrostatics with a daily-life example.
2. Explain Coulomb's law on the board with definitions and clear diagrams.
3. Discuss electric field and electric potential using simple, relatable examples.
4. Involve the students through short questions on capacitors.
5. Solve a numerical on Gauss's law together with the class.
6. Summarize the lesson and highlight the key points of Electrostatics.

Classroom Activities

• Group discussion: students compare electric field with electric potential.
• Board activity: solve a question on capacitors in pairs.
• Quick quiz on the key terms of Electrostatics.

Assessment Questions

1. Define Coulomb's law.
2. Differentiate between electric field and electric potential.
3. Solve a numerical based on capacitors.
4. Write a short note on Gauss's law.

Homework

• Read Electrostatics from the book and solve the exercise questions.
• Prepare short notes on Coulomb's law and electric field.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Current Electricity

Learning Objectives

• Define and explain Ohm's law in the context of Current Electricity.
• Differentiate between Kirchhoff's rules and Wheatstone bridge with suitable examples.
• Solve numerical problems related to potentiometer accurately.
• Apply the ideas of Current Electricity to everyday situations involving thermocouple.

Key Concepts

• Meaning, definition and scope of Ohm's law.
• Important properties, types and examples of Kirchhoff's rules.
• Relationship between Wheatstone bridge and potentiometer.
• Practical applications of Current Electricity, especially thermocouple.
• Key terms and a summary of the main points of Current Electricity.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Current Electricity with a daily-life example.
2. Explain Ohm's law on the board with definitions and clear diagrams.
3. Discuss Kirchhoff's rules and Wheatstone bridge using simple, relatable examples.
4. Involve the students through short questions on potentiometer.
5. Solve a numerical on thermocouple together with the class.
6. Summarize the lesson and highlight the key points of Current Electricity.

Classroom Activities

• Group discussion: students compare Kirchhoff's rules with Wheatstone bridge.
• Board activity: solve a question on potentiometer in pairs.
• Quick quiz on the key terms of Current Electricity.

Assessment Questions

1. Define Ohm's law.
2. Differentiate between Kirchhoff's rules and Wheatstone bridge.
3. Solve a numerical based on potentiometer.
4. Write a short note on thermocouple.

Homework

• Read Current Electricity from the book and solve the exercise questions.
• Prepare short notes on Ohm's law and Kirchhoff's rules.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Electromagnetism

Learning Objectives

• Define and explain magnetic field in the context of Electromagnetism.

• Differentiate between Ampere's law and galvanometer with suitable examples.

• Solve numerical problems related to voltmeter accurately.

• Apply the ideas of Electromagnetism to everyday situations involving cathode ray oscilloscope.

Key Concepts

• Meaning, definition and scope of magnetic field.

• Important properties, types and examples of Ampere's law.

• Relationship between galvanometer and voltmeter.

• Practical applications of Electromagnetism, especially cathode ray oscilloscope.

• Key terms and a summary of the main points of Electromagnetism.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Electromagnetism with a daily-life example.
2. Explain magnetic field on the board with definitions and clear diagrams.
3. Discuss Ampere's law and galvanometer using simple, relatable examples.
4. Involve the students through short questions on voltmeter.
5. Solve a numerical on cathode ray oscilloscope together with the class.
6. Summarize the lesson and highlight the key points of Electromagnetism.

Classroom Activities

• Group discussion: students compare Ampere's law with galvanometer.

• Board activity: solve a question on voltmeter in pairs.

• Quick quiz on the key terms of Electromagnetism.

Assessment Questions

1. Define magnetic field.
2. Differentiate between Ampere's law and galvanometer.
3. Solve a numerical based on voltmeter.
4. Write a short note on cathode ray oscilloscope.

Homework

• Read Electromagnetism from the book and solve the exercise questions.

• Prepare short notes on magnetic field and Ampere's law.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Electromagnetic Induction

Learning Objectives

• Define and explain Faraday's law in the context of Electromagnetic Induction.

• Differentiate between Lenz's law and mutual induction with suitable examples.

• Solve numerical problems related to transformers accurately.

• Apply the ideas of Electromagnetic Induction to everyday situations involving eddy currents.

Key Concepts

• Meaning, definition and scope of Faraday's law.

• Important properties, types and examples of Lenz's law.

• Relationship between mutual induction and transformers.

• Practical applications of Electromagnetic Induction, especially eddy currents.

• Key terms and a summary of the main points of Electromagnetic Induction.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Electromagnetic Induction with a daily-life example.
2. Explain Faraday's law on the board with definitions and clear diagrams.
3. Discuss Lenz's law and mutual induction using simple, relatable examples.
4. Involve the students through short questions on transformers.
5. Solve a numerical on eddy currents together with the class.
6. Summarize the lesson and highlight the key points of Electromagnetic Induction.

Classroom Activities

• Group discussion: students compare Lenz's law with mutual induction.

• Board activity: solve a question on transformers in pairs.

• Quick quiz on the key terms of Electromagnetic Induction.

Assessment Questions

1. Define Faraday's law.
2. Differentiate between Lenz's law and mutual induction.
3. Solve a numerical based on transformers.
4. Write a short note on eddy currents.

Homework

• Read Electromagnetic Induction from the book and solve the exercise questions.

• Prepare short notes on Faraday's law and Lenz's law.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: A.C. and Electronics

Learning Objectives

• Define and explain alternating current in the context of A.C. and Electronics.

• Differentiate between RLC circuits and diodes with suitable examples.

• Solve numerical problems related to rectifiers accurately.

• Apply the ideas of A.C. and Electronics to everyday situations involving transistors.

Key Concepts

• Meaning, definition and scope of alternating current.

• Important properties, types and examples of RLC circuits.

• Relationship between diodes and rectifiers.

• Practical applications of A.C. and Electronics, especially transistors.

• Key terms and a summary of the main points of A.C. and Electronics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce A.C. and Electronics with a daily-life example.
2. Explain alternating current on the board with definitions and clear diagrams.
3. Discuss RLC circuits and diodes using simple, relatable examples.
4. Involve the students through short questions on rectifiers.
5. Solve a numerical on transistors together with the class.
6. Summarize the lesson and highlight the key points of A.C. and Electronics.

Classroom Activities

• Group discussion: students compare RLC circuits with diodes.

• Board activity: solve a question on rectifiers in pairs.

• Quick quiz on the key terms of A.C. and Electronics.

Assessment Questions

1. Define alternating current.
2. Differentiate between RLC circuits and diodes.
3. Solve a numerical based on rectifiers.
4. Write a short note on transistors.

Homework

• Read A.C. and Electronics from the book and solve the exercise questions.

• Prepare short notes on alternating current and RLC circuits.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Dawn of Modern Physics

Learning Objectives

• Define and explain photoelectric effect in the context of Dawn of Modern Physics.
• Differentiate between Compton effect and de Broglie hypothesis with suitable examples.
• Solve numerical problems related to uncertainty principle accurately.
• Apply the ideas of Dawn of Modern Physics to everyday situations involving black body radiation.

Key Concepts

• Meaning, definition and scope of photoelectric effect.
• Important properties, types and examples of Compton effect.
• Relationship between de Broglie hypothesis and uncertainty principle.
• Practical applications of Dawn of Modern Physics, especially black body radiation.
• Key terms and a summary of the main points of Dawn of Modern Physics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Dawn of Modern Physics with a daily-life example.
2. Explain photoelectric effect on the board with definitions and clear diagrams.
3. Discuss Compton effect and de Broglie hypothesis using simple, relatable examples.
4. Involve the students through short questions on uncertainty principle.
5. Solve a numerical on black body radiation together with the class.
6. Summarize the lesson and highlight the key points of Dawn of Modern Physics.

Classroom Activities

• Group discussion: students compare Compton effect with de Broglie hypothesis.
• Board activity: solve a question on uncertainty principle in pairs.
• Quick quiz on the key terms of Dawn of Modern Physics.

Assessment Questions

1. Define photoelectric effect.
2. Differentiate between Compton effect and de Broglie hypothesis.
3. Solve a numerical based on uncertainty principle.
4. Write a short note on black body radiation.

Homework

• Read Dawn of Modern Physics from the book and solve the exercise questions.
• Prepare short notes on photoelectric effect and Compton effect.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Atomic Spectra

Learning Objectives

• Define and explain hydrogen spectrum in the context of Atomic Spectra.
• Differentiate between Bohr's model and X-rays with suitable examples.
• Solve numerical problems related to laser accurately.
• Apply the ideas of Atomic Spectra to everyday situations involving spectral series.

Key Concepts

• Meaning, definition and scope of hydrogen spectrum.
• Important properties, types and examples of Bohr's model.
• Relationship between X-rays and laser.
• Practical applications of Atomic Spectra, especially spectral series.
• Key terms and a summary of the main points of Atomic Spectra.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Atomic Spectra with a daily-life example.
2. Explain hydrogen spectrum on the board with definitions and clear diagrams.
3. Discuss Bohr's model and X-rays using simple, relatable examples.
4. Involve the students through short questions on laser.
5. Solve a numerical on spectral series together with the class.
6. Summarize the lesson and highlight the key points of Atomic Spectra.

Classroom Activities

• Group discussion: students compare Bohr's model with X-rays.
• Board activity: solve a question on laser in pairs.
• Quick quiz on the key terms of Atomic Spectra.

Assessment Questions

1. Define hydrogen spectrum.
2. Differentiate between Bohr's model and X-rays.
3. Solve a numerical based on laser.
4. Write a short note on spectral series.

Homework

• Read Atomic Spectra from the book and solve the exercise questions.
• Prepare short notes on hydrogen spectrum and Bohr's model.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Nuclear Physics

Learning Objectives

• Define and explain atomic nucleus in the context of Nuclear Physics.

• Differentiate between radioactivity and half-life with suitable examples.

• Solve numerical problems related to nuclear fission accurately.

• Apply the ideas of Nuclear Physics to everyday situations involving nuclear fusion.

Key Concepts

• Meaning, definition and scope of atomic nucleus.

• Important properties, types and examples of radioactivity.

• Relationship between half-life and nuclear fission.

• Practical applications of Nuclear Physics, especially nuclear fusion.

• Key terms and a summary of the main points of Nuclear Physics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Nuclear Physics with a daily-life example.
2. Explain atomic nucleus on the board with definitions and clear diagrams.
3. Discuss radioactivity and half-life using simple, relatable examples.
4. Involve the students through short questions on nuclear fission.
5. Solve a numerical on nuclear fusion together with the class.
6. Summarize the lesson and highlight the key points of Nuclear Physics.

Classroom Activities

• Group discussion: students compare radioactivity with half-life.

• Board activity: solve a question on nuclear fission in pairs.

• Quick quiz on the key terms of Nuclear Physics.

Assessment Questions

1. Define atomic nucleus.
2. Differentiate between radioactivity and half-life.
3. Solve a numerical based on nuclear fission.
4. Write a short note on nuclear fusion.

Homework

• Read Nuclear Physics from the book and solve the exercise questions.

• Prepare short notes on atomic nucleus and radioactivity.

Time Allocation

4–6 periods of 40 minutes each